Spectral line diagnostics – work split

M2 IRT, MC "Radiation" 2026–27

Goal. Tool that starts from a real spectrum, (1) fits the continuum, (2) finds and identifies the lines, (3) characterises them and derives the physical conditions of the gas. **Check-ins:** Oct 14, Nov 18. **Oral:** Dec 9 (50% of grade, may be individual).

Scope (updated after check-in 1, Oct 14). Optical emission-line spectra of *planetary nebulae*. Main target: NGC 7662 (HST/STIS G430L + G750L, MAST programme 12600, slit on the nebula); second target: NGC 7027 (programme 8285, richest red spectrum). The absorption work done before check-in 1 (Voigt profiles, equivalent widths, curves of growth, (N, b) fit, all validated on synthetic spectra) is kept as a separate module.

A – data → lines	**B – lines → physics**
• MAST download; 1D extraction from the 2D STIS image (sum the nebular rows, propagate errors) • Continuum fit and subtraction (sigma-clipping that rejects emission lines) • Nebular line list (H, He recombination lines; forbidden lines of O, N, S, Ne, Ar; air vs vacuum wavelengths) • Line detection and identification against the line list (radial velocity from the matched lines) • Interactive visualisation	• Gaussian line profiles including the instrumental resolution; multi-line fits of blends (Hα + [N II], Hγ + [O III] 4363) • Line fluxes with error propagation (noise and continuum) • Reddening from the Balmer decrement; dereddened fluxes • Electron temperature ([O III], [N II]), electron density ([S II], [O II]), ionic abundances • Uncertainties (Monte Carlo)

Interface. A outputs a `Spectrum` (flux, error, continuum) plus a line table: name, λ_{rest}, observed centre guess, fitting window. B only needs this and returns a table of line fluxes and the derived physical quantities.

Avoiding the dependency. B first works on *synthetic emission spectra* (known line fluxes on a known continuum) and on a hand-made line table for NGC 7662. A's automatic identification later produces the same table; B's code does not change.

Shared work. Repo and interfaces; synthetic spectrum generator; validation (inject known fluxes, check recovery; compare our temperatures and densities with PyNeb and with published values for the same nebulae); report and slides (each presents own half).

Timeline.

Until Oct 14	Repo, interfaces, synthetic generator; absorption chain (profiles, EW, COG) validated on synthetic spectra. MAST search for planetary nebula spectra.
Oct 14 – Nov 18	A: extraction of NGC 7662, continuum, line list, detection and identification. B: Gaussian fits, fluxes, reddening, electron temperature. Demo on the real spectrum.
Nov 18 – Dec 8	Electron density, abundances, uncertainties, NGC 7027 as second object, interactive viz, comparison with the literature, slides and rehearsal.
Dec 9	Oral presentation.

Working habits. Shared git repo, personal branches merged into `main` by reviewed pull requests; 15–20 min weekly sync; integrate A's output into B's fits every 1–2 weeks; review each other's code so both understand the full pipeline.

Status (see README progress log for details). Done: `core` data classes, `profiles` (Gaussian and Voigt optical depth), `ew`, `io.make_synthetic`, `linelist` (CIV, SiII, FeII), `cog`, with validation notebooks 00–04. Candidate planetary nebula spectra found in MAST; the pipeline 1D products are too noisy for an extended nebula, so we extract our own. Next: `io` reader and extraction for STIS `_sx2` files, then `continuum`.